

Canada's Magnificent Condo

Testing reveals how well this building's design and construction are withstanding the test of time—and building occupancy.

by Duncan Hill

How do you know if a building that was designed to be innovative, energy efficient, and healthy is fulfilling that design? You test it after the occupants have moved in. In 1998–99, a six-story, 48-unit condominium was designed and built to meet the four main goals of Canada Mortgage and Housing Corporation's (CMHC) Innovative Durable Energy Efficient Affordable Solutions (IDEAS) Challenge, and Natural Resources Canada's Commercial Building Incentive program (CBIP). These goals include envelope durability, energy and water efficiency, indoor air quality (IAQ), and environmental and resource conservation. The building was designed to consume 35% less energy than a similar building that meets the Canadian Model National Energy Code for Buildings, and to provide better thermal comfort and IAQ than are found in typical apartment buildings.

Many advanced or innovative features were integrated into the design and construction of the building:

The building has a tight and well-insulated envelope consisting of an exterior insulating finish system (EIFS) and brick veneer cladding. The EIFS system consists of stucco applied over a semirigid insulation that is fastened through the wall sheathing to 4-inch steel studs filled with R-8 fiberglass batt insulation.

Thermal bridging through the building envelope by floor slab brick ledges and balconies was eliminated. The supporting brick ledges were projected from the slab by structural brackets so that insulation could be placed between the brick ledge and the slab edge (see Figure 1).



Windows are low-e, spectrally selective, and argon filled, with insulating spacers and fiberglass frames. Spectrally selective double glazing with visible transmittance of 0.72, a U-value of 9 Btu/ft²•hr•°F (1.6 W/m²•°C), and a solar heat gain coefficient of 0.42 was installed to minimize overheating in the summer. The building has individual metering for natural gas and electricity.

Natural gas-fired combination space and water heating was installed in each suite. The 40-gallon (151-liter) hot water tanks (with an energy factor, or EF, of 0.58) are connected to site-built fan coil units. The in-suite fan coil units

utilize variable-speed, high-efficiency electronically commutated motors (ECMs) for low energy consumption (see "The Electric Side of Gas Furnaces," *HE* Nov/Dec '03, p. 24).

The building has individual in-suite and separate common-area heat recovery ventilation (HRV) systems. These systems have a typical capacity of 60 CFM (30 liters per second; L/s), or the capacity to provide 0.35 ACH. They have a high-speed capacity of 120 CFM (60 L/s), and an average heat recovery efficiency of 60%.

The building has corridor air heat recovery ventilation (100 CFM, or 50 L/s delivered to each corridor to provide 1.6 ACH). The ventilation system uses the garbage chute as an exhaust air plenum.

Water-efficient appliances and fixtures in the condo include 1.6 gallon (6 liters) per flush toilets; aerators on faucets; and low-flow showerheads (that provide less than 2.2 gallons or 8 liters per minute at 60 PSI).

The building also has energy-efficient appliances and lighting in common areas and parking garages. (The T-8 and compact fluorescent lighting in the corridors and parking garages uses 30% less energy than is specified in the Canadian Model National Energy Code.)

The developer maintains that this building probably cost no more to construct than the building he would otherwise have constructed for this clientele.

Research Program

CMHC initiated a research project to evaluate the performance of the building in terms of energy and water use, indoor environment, and building envelope

durability. Enermodal Engineering Limited, the project consultant, developed and undertook an extensive monitoring and evaluation program.

Building Envelope

Enermodal developed a program to ensure that the building envelope was designed and constructed to meet minimum leakage criteria of 0.197 CFM/ft²

mechanical, electrical, and cladding support; and wall-roof interfaces. Once the air leakage details were defined, a quality assurance plan was developed. This plan involved trades training and site inspection. Enermodal performed the airtightness testing in cooperation with Can Am Building Envelope Specialists. The testing focused on isolated wall areas, entire apartments, and finally, the complete building envelope.

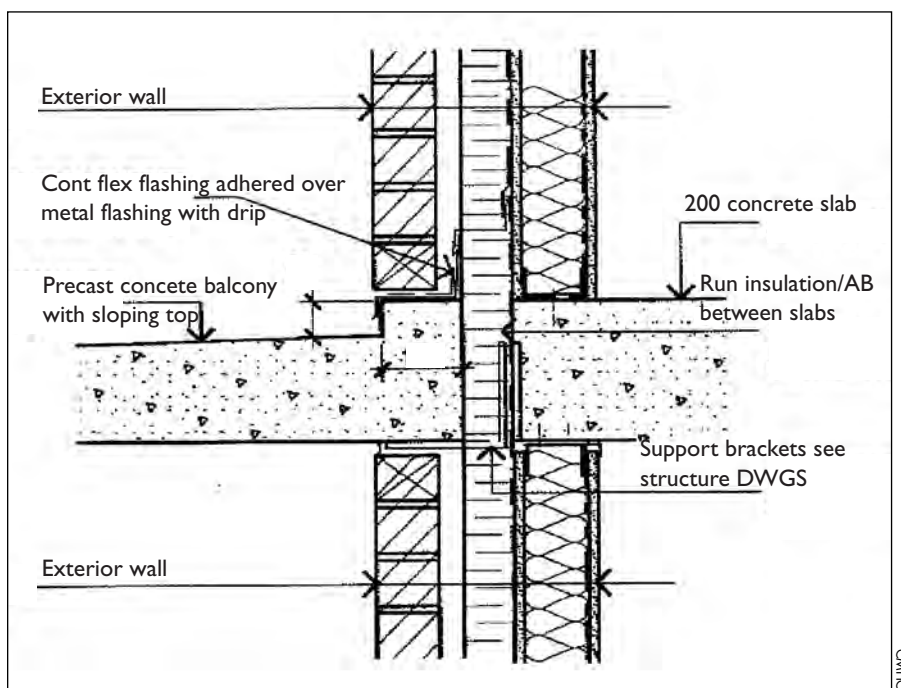


Figure 1. Structural brackets that support the brick ledge provide a space between the ledge and the floor slab where insulation is applied.

(1.0 L/s per square meter) at 75 Pa. (CMHC uses 75 Pa in order to ensure that tests are done at well above any naturally occurring pressures. This is the rating point for both air barrier material and air barrier systems in the Canadian Model National Building Code.) This criterion was selected because it is roughly one-third to one-half of the air leakage rate of the typical multifamily residential building (see "Valuing Air Barriers," *HE* Sept/Oct '01, p. 29).

Enermodal evaluated the building plans and identified areas where air sealing details had to be developed. These include window-wall interfaces; wall-floor intersections; penetrations for

and domestic hot water tank (used in one suite), and the boilers.

The energy and water monitoring was used to determine monthly and yearly consumption. This information helped Enermodal to calculate the energy performance of combination space- and water-heating systems used in multifamily buildings. The energy data were also used to compare the performance of the building to that of other multifamily buildings.

Ventilation System

The performance of the in-suite HRV systems was assessed via flow measurements and perfluorocarbon tracer (PFT) tests from Brookhaven National Laboratory, and short-term tracer gas decay tests (instantaneous SF₆). The objective of the tests was to determine the ability of the HRV systems to exchange, distribute, and circulate ventilation air in the apartments. The performance of the parking garage and corridor ventilation systems was also assessed via tracer gas tests. The objective was to determine whether the system effectively ventilated these zones, and whether it prevented air from leaking from these zones to other areas of the building.

Indoor Environment

The indoor environment in one suite was monitored for airborne particulates, formaldehyde, and volatile organic compounds (VOCs). Long-term monitoring of temperature, relative humidity (RH), and CO₂ was also performed in three suites. The goal of these tests was to determine if the measures taken to enhance IAQ were successful.

Wall Assemblies

The temperature, moisture, and pressure regimes through the brick veneer and EIFS wall sections were monitored continuously to determine if conditions in the wall assemblies were conducive to long term durability. The monitoring

Energy and Water Consumption

The natural gas, electricity, and water consumption for the common areas and suites was monitored via the utility meters for the first year post occupancy. (The study did not include data from occupants' behavior because occupancy rates and thermostat settings were not part of the monitoring plan.) Detailed energy consumption patterns were monitored continuously in three suites and a common area to characterize the performance of the combination space- and water-heating systems, the high-efficiency furnaces

program was conducted after the building had been fully occupied for one year. Most of the monitoring was done automatically with on-site instrumentation and data-recording equipment.

Findings

The building consumed 12.7 kWh/ft² (137 kWh per square meter) for the year monitored. (Building consumption is natural gas and electricity expressed as equivalent kWh.) While this consumption exceeded the performance target of 11.6 kWh/ft² (125 kWh per square meter), it is still far below the Model National Energy Code for Buildings value of 18.1 kWh/ft² (195 kWh per square meter) and the typical annual energy consumption of multifamily buildings, which can exceed 28 kWh/ft² (300 kWh per square meter). (See Figures 1 and 2 for the distribution of annual energy use in the building, and a comparison of predicted and actual energy use.)

The seasonal energy efficiency of the combination space- and water-heating systems was found to range between 43% and 63%. The low efficiency was thought to be a function of inadequate commissioning practices, such as measuring and adjusting the combustion efficiency, and low space-heating and domestic hot water loads.

The average electrical baseload per suite was 11.8 kWh per day. When the space-cooling load is included, the average electrical load was 15 kWh per day. The average natural gas use, for space heating and domestic hot water, averaged 92 ft³ (2.6 cubic meters) per day per suite. The central cooling system consumed 61,610 kWh during the monitored year.

Water consumption, metered centrally, was 2,719 ft³ (77 cubic meters) per suite per year. This represents very

little water use compared to other multifamily buildings, which typically use more than 7,600 ft³ (215 cubic meters) per unit per year. The low consumption is probably due to the water saving appliances—and to the fact that the occupants are affluent seniors who can spend the winter months away.

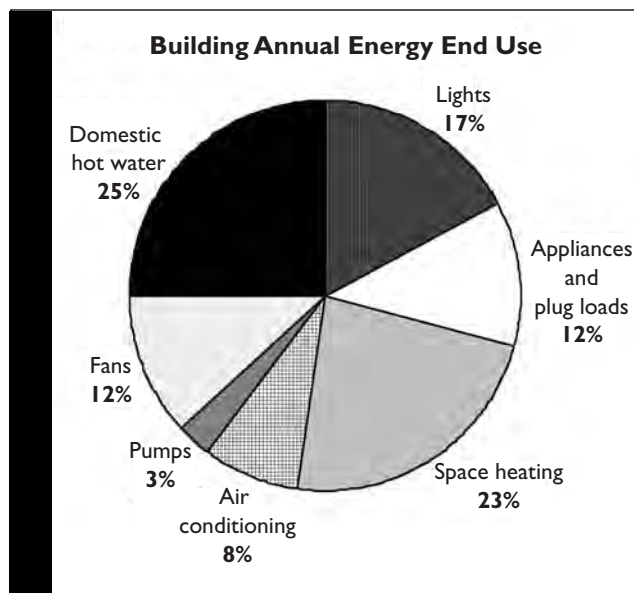


Figure 1. The relatively low energy efficiency of the combination space- and water-heating systems was likely due to inadequate commissioning of the systems, and low loads.

The building envelope airtightness target was met despite the challenges of integrating air leakage control procedures across the many stages of design and construction and the many different trades that must be convinced to buy into the process. The improved building envelope no doubt contributed to occupant comfort and low space-heating energy consumption. The level of airtightness achieved in the condominium was 0.24 CFM/ft² (1.2 L/s per square meter) at 75 Pa. This represents a reduction of 75% over conventional buildings, where measured airtightness has been found to range from 0.17 CFM/ft² (0.83 L/s per square meter) at 75 Pa to 2 CFM/ft² (10 L/s per square meter) at 75 Pa—with a mean airtightness of 0.76 CFM/ft² (3.8 L/s per square meter) at 75 Pa.

Temperature, RH, and air pressure were monitored at several points through the building envelope, in various locations. The monitoring of the wall assemblies showed that the condensation potential within the walls was low. The combination of temperature and moisture conditions required for

condensation to form within the wall assemblies was unlikely to occur. This should help to ensure the long-term durability of the wall structures.

The in-suite HRVs were found to operate effectively. The tracer gas tests showed that ventilation air was well distributed and well circulated in each apartment—something that is rarely achieved in conventional apartment buildings. The testing also showed that the HRVs could distribute and circulate ventilation air with the forced-air system fan turned off, thereby saving electricity. The ventilation system that delivered air to the corridors and exhausted air from the garbage chute was found to help contain garbage chute odors, prevent parking garage gases from leaking into the apartments, and maintain acceptable air quality in the corridors of each floor.

The air quality monitoring in one apartment found common contaminants to be within Health Canada guidelines. The airborne respirable particles were measured at 2.5 µg per cubic meter. Health Canada guidelines require that these particles be kept below 40 µg per cubic meter for long-term exposure. Formaldehyde was measured at less than 0.033 ppm; Health Canada guidelines are 0.05 ppm. VOCs were below detection. Indoor air temperatures in three apartments were held to a fairly constant average of 74°F (23.4°C) with a standard deviation of 2°F–3°F (1°C–1.9°C) during the space-heating and -cooling seasons. Indoor RH in the three apartments varied from 15% to 70%, but average summertime RH

ranged from 40% to 60%, while the average winter range was 25% to 40% (condo owners decide individually whether or not to use humidifiers). Carbon dioxide readings averaged 516–827 ppm (the ASHRAE recommendation is less than 1,000 ppm), but with occasional readings of up to 2,000 ppm. The higher readings are thought to have occurred when occupants deactivated the HRVs. The in-suite HRVs appeared to maintain good IAQ.

Implications for the Housing Industry

The verified performance of this innovative multifamily building indicates that it is possible to include many advanced and environmentally responsible features in a fairly conventional construction

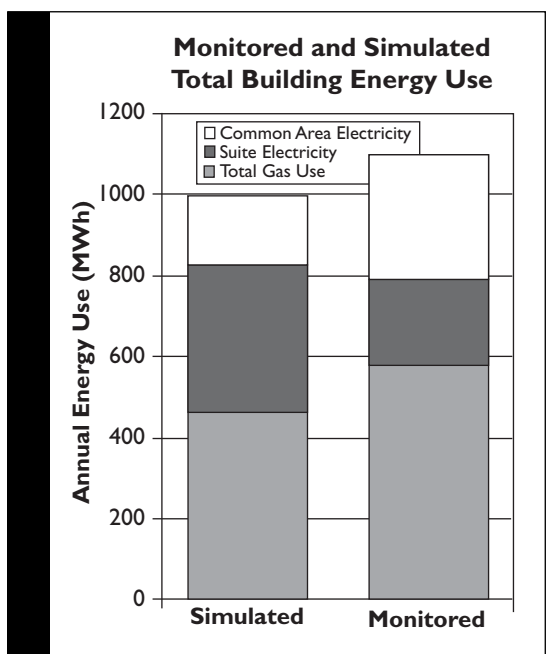


Figure 2. While the building energy consumption exceeded the performance target, the building used far less energy than the Model National Energy Code for Buildings.

project. The integration of an air leakage control procedure into the construction, design, and commissioning process is perhaps one of the most important aspects of this project. The in-suite and corridor ventilation strategies represent significant departures from conventional designs, but demonstrate significantly improved performance.

The study also shows that the energy use—hence the environmental impact—of high-end multifamily buildings can be significantly reduced through thoughtful design and construction techniques without risking the commercial viability of the project.

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